Laplace's equation

With partial differential equations, everything we've done so far has been for the one-dimensional heat equation.

Now we want to look at a two-dimensional partial differential equation: Laplace's equation.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\nabla^2 u = 0$$

The ∇^2 operator is the **Laplacian**, and it tells us to take the second derivative with respect to each variable in the function, then add those second derivatives. So if u is a function in x and y, $u(x, y)$, then the Laplacian of u, $\nabla^2 u$, will be the sum of the second derivative with respect to x and the second derivative with respect to y.

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

Where previously we only had temperature change along one dimension (from left to right, and/or from right to left), now we can look at temperature change in two dimensions (from left to right, and/or from right to left, but also from top to bottom, and/or from bottom to top, at the same time).

© Krista King Math



The way we solve Laplace's equation depends on the geometry of the system, so we'll start with the simplest scenario, a two-dimensional rectangular region.

A rectangular region

Let's imagine that we have a rectangle with length defined on $0 \leq x \leq L$, and height defined on $0 \leq y \leq H$. Then we can define boundary conditions for Laplace's equation along each boundary of the rectangle.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$u(0,y) = g_1(y) \qquad u(L,y) = g_2(y)$$

$$u(x,0) = f_1(x) \qquad u(x,H) = f_2(x)$$

To solve this boundary value problem, we'll actually solve it four times, generating four solutions $u_1(x,y)$, $u_2(x,y)$, $u_3(x,y)$, and $u_4(x,y)$. Each time, we'll keep one boundary condition nonhomogeneous while we hold the other three homogeneous.

1. $u_1(x,0) = f_1(x)$ $\quad u_1(0,y) = 0$ $\quad u_1(x,H) = 0$ $\quad u_1(L,y) = 0$
2. $u_2(x,0) = 0$ $\quad u_2(0,y) = g_1(y)$ $\quad u_2(x,H) = 0$ $\quad u_2(L,y) = 0$
3. $u_3(x,0) = 0$ $\quad u_3(0,y) = 0$ $\quad u_3(x,H) = f_2(x)$ $\quad u_3(L,y) = 0$
4. $u_4(x,0) = 0$ $\quad u_4(0,y) = 0$ $\quad u_4(x,H) = 0$ $\quad u_4(L,y) = g_2(y)$

Once we find all four solutions, then solution to Laplace's equation will be

© Krista King Math

$$u(x,y) = u_1(x,y) + u_2(x,y) + u_3(x,y) + u_4(x,y)$$

To find each of the four solutions, we'll use separation of variables and the product solution, along with the three homogeneous boundary conditions, then we'll plug in the single nonhomogeneous boundary condition to the result that we get.

For instance, to find $u_2(x,y)$ we have to solve the boundary value problem

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u_2(0,y) = g_1(y) \qquad u_2(L,y) = 0$$

$$u_2(x,0) = 0 \qquad u_2(x,H) = 0$$

Using the product solution $u_2(x,y) = v(x)w(y)$, the partial differential equation can be broken into two ordinary differential equations,

$$\frac{d^2v}{dx^2} - \lambda v = 0 \qquad \frac{d^2w}{dy^2} + \lambda w = 0$$

$$v(L) = 0 \qquad w(0) = 0 \qquad w(H) = 0$$

The solution to the second ordinary differential equation boundary value problem is

$$\lambda_n = \left(\frac{n\pi}{H}\right)^2 \qquad w_n(y) = \sin\left(\frac{n\pi y}{H}\right) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the first equation, we get

© Krista King Math

$$\frac{d^2v}{dx^2} - \left(\frac{n\pi}{H}\right)^2 v = 0$$

$$v(L) = 0$$

Because the coefficient on v is positive, we know that the solution to this second order equation will be

$$v(x) = c_1 e^{\frac{n\pi x}{H}} + c_2 e^{-\frac{n\pi x}{H}}$$

which can also be rewritten as

$$v(x) = c_1 \cosh\left(\frac{n\pi x}{H}\right) + c_2 \sinh\left(\frac{n\pi x}{H}\right)$$

Equivalently, this hyperbolic form can be rewritten as

$$v(x) = c_1 \cosh\left(\frac{n\pi(x-L)}{H}\right) + c_2 \sinh\left(\frac{n\pi(x-L)}{H}\right)$$

which allows us to apply $v(L) = 0$. When we do, we find $c_1 = 0$, and the solution is

$$v(x) = c_2 \sinh\left(\frac{n\pi(x-L)}{H}\right)$$

Then the product solution is

$$u_2(x, y) = v(x)w(y)$$

$$u_2(x, y) = c_2 \sinh\left(\frac{n\pi(x-L)}{H}\right) \sin\left(\frac{n\pi y}{H}\right)$$

© Krista King Math



$$u_2(x,y) = B_n \sinh\left(\frac{n\pi(x-L)}{H}\right)\sin\left(\frac{n\pi y}{H}\right) \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$u_2(x,y) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi(x-L)}{H}\right)\sin\left(\frac{n\pi y}{H}\right)$$

Substitute $u_2(0,y) = g_1(y)$.

$$u_2(0,y) = g_1(y) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi(-L)}{H}\right)\sin\left(\frac{n\pi y}{H}\right)$$

This is just like a Fourier sine series, except that the coefficients are more complicated.

$$B_n \sinh\left(\frac{n\pi(-L)}{H}\right) = \frac{2}{H}\int_0^H g_1(y)\sin\left(\frac{n\pi y}{H}\right)\,dy \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$B_n = \frac{2}{H\sinh\left(\frac{n\pi(-L)}{H}\right)}\int_0^H g_1(y)\sin\left(\frac{n\pi y}{H}\right)\,dy \qquad n = 1,\ 2,\ 3,\ \ldots$$

Using a similar method, we can find the solutions for all four functions, $u_1(x,y)$, $u_2(x,y)$, $u_3(x,y)$, and $u_4(x,y)$ to be

$$u_1(x,y) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi(y-H)}{L}\right)\sin\left(\frac{n\pi x}{L}\right)$$

$$B_n = -\frac{2}{L\sin\left(\frac{n\pi H}{L}\right)}\int_0^L f_1(x)\sin\left(\frac{n\pi x}{L}\right)\,dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

© Krista King Math



$$u_2(x,y) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi(x-L)}{H}\right)\sin\left(\frac{n\pi y}{H}\right)$$

$$B_n = -\frac{2}{H\sinh\left(\frac{n\pi L}{H}\right)}\int_0^H g_1(y)\sin\left(\frac{n\pi y}{H}\right)\,dy \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$u_3(x,y) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi y}{L}\right)\sin\left(\frac{n\pi x}{L}\right)$$

$$B_n = \frac{2}{L\sin\left(\frac{n\pi H}{L}\right)}\int_0^L f_2(x)\sin\left(\frac{n\pi x}{L}\right)\,dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$u_4(x,y) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi x}{H}\right)\sin\left(\frac{n\pi y}{H}\right)$$

$$B_n = \frac{2}{H\sinh\left(\frac{n\pi L}{H}\right)}\int_0^H g_2(y)\sin\left(\frac{n\pi y}{H}\right)\,dy \qquad n = 1,\ 2,\ 3,\ \ldots$$

Therefore, the solution to Laplace's equation over the rectangular domain $0 \le x \le L$ and $0 \le y \le H$ is given by,

$$u(x,y) = u_1(x,y) + u_2(x,y) + u_3(x,y) + u_4(x,y)$$

where the values of $u_1(x,y)$, $u_2(x,y)$, $u_3(x,y)$, and $u_4(x,y)$ are defined above.

Let's do another example where we derive one of these $u_n(x,y)$ solutions, just to make sure we have a good feel for the process.

Example

© Krista King Math

Use the product solution to find $u_1(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq L$ and $0 \leq y \leq H$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u_1(x,0) = f_1(x) \qquad u_1(0,y) = 0$$

$$u_1(x, H) = 0 \qquad u_1(L, y) = 0$$

If we find the first and second derivatives of the product solution,

$$u(x, y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d^2 v}{dx^2}w \qquad \frac{\partial^2 u}{\partial y^2} = v\frac{d^2 w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2 v}{dx^2}w + v\frac{d^2 w}{dy^2} = 0$$

then we can separate variables.

$$\frac{d^2 v}{dx^2}w = -v\frac{d^2 w}{dy^2}$$

© Krista King Math

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} = \lambda w(y)$$

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

Our boundary conditions are $w(H) = 0$, $v(0) = 0$, and $v(L) = 0$, and the solution to the first ordinary differential equation boundary value problem is

© Krista King Math



$$\lambda_n = \left(\frac{n\pi}{L}\right)^2 \qquad v_n(x) = \sin\left(\frac{n\pi x}{L}\right) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for λ into the second equation, we get

$$\frac{d^2w}{dy^2} - \left(\frac{n\pi}{L}\right)^2 w = 0$$

$$w(H) = 0$$

Because the coefficient on w is negative, we know that the solution to this second order equation will be

$$w(y) = c_1 \cosh\left(\frac{n\pi y}{L}\right) + c_2 \sinh\left(\frac{n\pi y}{L}\right)$$

Equivalently, this can be rewritten as

$$w(y) = c_1 \cosh\left(\frac{n\pi(H-y)}{L}\right) + c_2 \sinh\left(\frac{n\pi(H-y)}{L}\right)$$

which allows us to apply $w(H) = 0$. When we do, we find $c_1 = 0$, and the solution is

$$w(y) = c_2 \sinh\left(\frac{n\pi(H-y)}{L}\right)$$

Then the product solution is

$$u_1(x, y) = v(x)w(y)$$

$$u_1(x, y) = c_2 \sinh\left(\frac{n\pi(H-y)}{L}\right) \sin\left(\frac{n\pi x}{L}\right)$$

© Krista King Math



$$u_1(x,y) = B_n \sinh\left(\frac{n\pi(H-y)}{L}\right)\sin\left(\frac{n\pi x}{L}\right) \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$u_1(x,y) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi(H-y)}{L}\right)\sin\left(\frac{n\pi x}{L}\right)$$

Substitute $u_1(x,0) = f_1(x)$.

$$u_1(x,0) = f_1(x) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi H}{L}\right)\sin\left(\frac{n\pi x}{L}\right)$$

This is just like a Fourier sine series, except that the coefficients are more complicated.

$$B_n \sinh\left(\frac{n\pi H}{L}\right) = \frac{2}{L}\int_0^L f_1(x)\sin\left(\frac{n\pi x}{L}\right)\,dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$B_n = \frac{2}{L\sinh\left(\frac{n\pi H}{L}\right)}\int_0^L f_1(x)\sin\left(\frac{n\pi x}{L}\right)\,dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

© Krista King Math